

JOSE EMMANUEL RODRIGUEZ RIOS

COMPUTER SCIENTIST | AI/HPC | CYBERSECURITY

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PROFESSIONAL PROFILE

Computer scientist and cybersecurity engineer whose work connects AI, high-performance computing, secure research software, and network defense. National-laboratory experience spans regulatory-genomics training on leadership-class systems at Argonne and containerized cyber-research infrastructure at Lawrence Livermore; community work includes an energy-billing application for a solar cooperative in Puerto Rico.

Research interests: AI for scientific computing; research software engineering; secure distributed systems; vulnerability assessment; cybersecurity education

EDUCATION

Ph.D., Software Engineering | University of Puerto Rico - Mayagüez

Aug 2026 - Present

M.S., Computer Science | University of Puerto Rico - Río Piedras

Aug 2022 - May 2025 | GPA: 4.0/4.0

B.S., Computer Science | University of Puerto Rico - Río Piedras

Aug 2015 - Dec 2020 | Minor: Cybersecurity

TECHNICAL SKILLS

Programming: Python, C, C++, Java, Assembly

AI/ML & Data: PyTorch, TensorFlow, Keras, Hugging Face, LLMs/transformers, Pandas, NumPy, Matplotlib, Jupyter

High-Performance Computing: SLURM, GPU computing, NERSC Perlmutter, ALCF Polaris, ALCF Aurora, Jupyter-to-batch workflows

Vulnerability assessment: Nmap, OpenVAS, Vulners, vulscan, Nikto, Nuclei

Reverse engineering: Ghidra, BSim, GDB

Digital forensics: TestDisk, Foremost, Autopsy

Security testing: Metasploit, Burp Suite, Wireshark, Scapy, Hashcat, Hydra, Gobuster, John the Ripper

DevOps & Platforms: Linux, Docker, GitLab CI/CD and runners, Git/GitHub, Elasticsearch, Ollama; familiar with Kubernetes and OpenShift

Networking & VPN: OpenVPN, WireGuard, client-server deployments

Web, Data & Automation: Flask, Dash, Streamlit, Playwright, MySQL, SQLite, openpyxl

Operating Systems: Linux, Windows, Kali Linux

Languages: English, Spanish

INDUSTRY AND NATIONAL LABORATORY EXPERIENCE

Research Intern | Argonne National Laboratory | Lemont, IL

Jun 2026 - Aug 2026

Project: Leveraging Large Language Models for Regulatory Genomics on HPC

- Developed and tested an educational DNA sequence-classification workflow with **PyTorch**, **Hugging Face**, and **DNABERT-style transformers** on **NERSC Perlmutter** and **ALCF Polaris** and Aurora, spanning **CPU** baselines and **GPU**-accelerated training.
- Compared **6-mer** and **byte-pair-encoding (BPE)** tokenization for DNA: similar accuracy (**81%** vs. **80%**) while BPE cut tokens per sequence from **195** to **34**.
- Designed fixed-configuration **CPU**-versus-**GPU** experiments to isolate acceleration benefits, then scaled workloads to demonstrate **GPU**-enabled capability.
- Created **Jupyter** lessons and diagrams covering transformer architectures, embeddings, classification pipelines, and **SLURM** commands including **sbatch**, **squeue**, and **sacct**.
- Led a **10-student** team within the **87-student** bootcamp as project lead: scoped the workflow to a **10-12-hour** project and coached two sub-teams through their final presentations to peers and laboratory staff.
- Guided one sub-team to train a DNA transformer from scratch, compare five tokenizations, and reach a validation **AUROC of 0.887** with overlapping **6-mers**.
- Guided the second sub-team to fine-tune **DNABERT** as a **CTCF**-binding classifier on **K562** data, reaching **91.7%** evaluation accuracy (**99.1%** training) after about **17 minutes** of training, with external testing on an astrocyte **CTCF** dataset.
- Authored a first-time project-lead guide and reusable project template to onboard future interns.

DevOps Intern | *Lawrence Livermore National Laboratory* | Livermore, CA

Jun 2024 - Aug 2024

- Containerized cyber-research applications and integrated **GitLab**-based **CI/CD** pipelines into cyber research workflows in an Agile environment.
- Built an **Elasticsearch** container for **Ghidra BSim**-extracted binary features, enabling security researchers to query malware-like samples at scale.
- Deployed a **Docker**-hosted **Ollama (Llama 3)** large language model service for the **PaperQA** PDF-reading application over LLNL research papers.

Cybersecurity Intern | *GM Sectech* | San Juan, PR

Jun 2022 - Aug 2022

- Collaborated with penetration testers on credential-theft, password-cracking, and domain-compromise scenarios; trained on **Metasploit**, **Hashcat**, and **Burp Suite**.
- Researched and executed internal attack techniques including phishing and malicious USB delivery for awareness training.
- Developed a **Python/Twilio** smishing (SMS phishing) simulator that sent controlled SMS lures and captured user interaction for security education.

RESEARCH AND ACADEMIC EXPERIENCE

Graduate Research and Teaching Assistant

Aug 2022 - May 2025

Laboratory of Dr. José Ortiz Ubarri and Department of Computer Science, UPR Río Piedras | San Juan, PR

- Designed and implemented **NVSRCO**, a network situational-awareness and vulnerability-scanning platform for organizations with limited security budgets, built entirely on open-source components so the only cost is the analysis server.
- Combined host discovery, port and service enumeration, vulnerability assessment, persistent scan storage, and secure web reporting in an end-to-end system that scans every host on a network; each vulnerability scan takes about **30 minutes**.
- Integrated **Python**, **Nmap**, and **OpenVAS** into a remote scanning workflow. The published implementation uses **CSV** reports, a **Dash** dashboard, **WireGuard** connectivity, and **Docker** configurations.
- Led laboratory sessions for Operating Systems, Computer Architecture, and Cybersecurity courses, training undergraduates in **Linux**, **Ghidra**-based assembly inspection, buffer overflow discovery, exploit development, and packet analysis with **Wireshark** and **Scapy**.
- Authored Capture the Flag (CTF) challenges in reverse engineering, cryptography, web, and forensics with progressive difficulty, and ran coding and outreach events for about **60** high-school students from Puerto Rico.

Undergraduate Research Assistant

2020

Laboratory of Dr. José Ortiz Ubarri, UPR Río Piedras | San Juan, PR

- Developed a deception-based defense project: a decoy university website using honeypot techniques that drew about **1,000** automated attack attempts per day and tracked where the attacks originated.

Undergraduate Software Assistant

2019

Molecular Biology Laboratory of Dr. José A. Rodríguez-Martínez, UPR Río Piedras | San Juan, PR

- Maintained and extended **NRLB**, a **Java** tool for analyzing large-scale protein-DNA interaction data from sequencing experiments supported by **NIH SC1GM127231**.
- Adapted analysis code for specific DNA-binding sequences and provided software support to laboratory researchers.

COMMUNITY AND LEADERSHIP

Volunteer Software Developer | *Cooperativa de Energía La Margarita (Abeyno Coop)* | Salinas, PR

2024 - Present

- Built Solar Fleet Reporter, a local **Python/Streamlit** application (about **1,700** lines, **88** offline test checks) that pulls monthly energy data for the cooperative's solar fleet from **Sol-Ark**, **FranklinWH**, and **sonnen** portals through their APIs and **Playwright**-driven logins.
- Implemented the cooperative's net-metering billing rules (three billing cases, per-site rates), **CSV** and **Excel** export, printable homeowner statements, and offline-site alerts so members can be billed accurately each month.

Peer Mentor | *Intro to HPC Bootcamp, Argonne National Laboratory*

Aug 2025

- Mentored a **5-student** team (one of two peer mentors on a **10-student** project) applying **Python** and data science to design an energy-efficient HPC system for fusion workloads under a **12 MW** power cap.

SELECTED PROJECTS

Network Vulnerability and Situational Awareness Scanner (NVSRCO)

Aug 2022 - Dec 2025

Graduate research project

Project: Source and report | github.com/joserico00/NVSRCO | repositorio.upr.edu/handle/11721/4342

- Scanner built on open-source components and described under Research and Academic Experience; the technical report is published in the UPR institutional repository.

Deep Learning for Malicious Traffic Detection | Coursework and continued development

2024 - 2026

- Developed **TensorFlow/Keras** experiments on **IoT-23** network-connection logs; the current pipeline splits by capture and fits preprocessing only on training data to prevent leakage.
- Evaluates a **class-weighted dense neural network** against a **logistic-regression** baseline using balanced accuracy, precision, recall, F1, ROC AUC, and PR AUC; earlier model runs and their limitations are documented separately.

Malware Image Classification (Maling) | Deep-learning project

- Trained a **TensorFlow/Keras** convolutional neural network on the **Maling** dataset (**9,401** grayscale malware images, **25 families**) to **97.7%** validation accuracy.

Operating Systems Projects | UPR Río Piedras

2022

Project: Source | github.com/joserico00/Operating-System-Projects

- Implemented **FIFO**, **Working Set Clock**, and **Optimal** page-replacement algorithms over a simulated virtual-memory list in **Python**.
- Modeled producer-consumer coordination with sockets, threads, and mutex-protected critical sections.
- Built a distributed file system with metadata and datanode servers supporting list and copy operations over sockets.

University Enrollment Database | UPR Río Piedras

2019

- Developed a normalized **MySQL** database modeling course enrollment, professors, courses, and students with unique identifiers.

ALCF Compute Finder

2026

- Built a **Streamlit** workload-to-system recommendation tool with explainable rankings, hard-requirement checks, job-script guidance, and a browser-based **stlite** option; unknown specifications never satisfy a requirement.

1UPPER Game Catalog Enrichment Pipeline

2026

- Built resumable **Python** ingestion and enrichment scripts for a software-engineering project, combining **IGDB** metadata and playtime information into CSVs for database import; implemented retries, checkpoints, caching, and offline tests.

Reporter Assay Analysis

2026

- Built a notebook workflow that reshapes luciferase-assay spreadsheets into tidy data, compares conditions against a baseline, and produces figures; included a synthetic example workbook.

Epidemic Modeling Teaching Notebooks

2026

- Developed **NumPy** teaching workflows for deterministic **SIR** models and stochastic outbreak simulations, with two student tracks, a master notebook, reference metrics, and HTML previews.

yt2ogg Audio Loop Tool

2026

- Built a shared desktop/CLI pipeline for seamless Ogg loops using **NumPy** beat tracking, **FFmpeg** audio processing, and standalone executable packaging with self-tests.

WORKSHOPS AND OUTREACH

- Intro to HPC Bootcamp Participant | Berkeley Lab / NERSC** **2023**
- Worked with a team on heatwave and power-outage analysis using **Python**, **pandas**, **Matplotlib**, and **Perlmutter**; presented results in a lightning talk. Power-outage notebooks.
- Cybersecurity Workshop Developer | UPR Río Piedras** **2023 - 2025**
- Created reusable CTF challenge generators and walkthroughs covering cryptography, forensics, **Linux**, scripting, and related security topics. CTF workshop materials.
- Volunteer | Argonne 80th Anniversary Open House** **2026**
- Helped run a human neural-network demonstration and built an LLM-assisted version to support single-visitor participation.

COMPETITIONS AND HONORS

- 3rd Place (25 teams) - AI/ML Hackathon | Lockheed Martin | Mayagüez, PR** **Sep 2026**
- Built the team's final entry, a weighted-boxes-fusion ensemble of four transformer detectors (**RT-DETR-L**, **Grounding DINO Base**, and two **RF-DETR-L** models) that locates plastic bags in waste-truck camera images from the Remondis Contamination Dataset, scoring **0.8441 mAP@50** on the organizers' hidden test set.
 - Raised validation mAP@50 from the **0.61** starter baseline to **0.89** over **40+** logged experiments on **AWS SageMaker** (NVIDIA A10G GPU), adopting a change only when it gained at least 0.005 and won 95% of 1,000 paired bootstrap draws.
- 1st Place (21 teams) - Data Analytics Challenge | CAHSI** **2026**
- Built a **LightGBM** network-intrusion classifier, achieving **0.9428 Macro-F1**. Classified benign traffic and **35 attack types** using approximately **1.7 million** training flows, with leakage-safe validation and class-imbalance handling.
- 2nd Place, Advanced Team Category - Cyber Hackathon | CAHSI / GMIS** **2026**
- Solved penetration-testing, reverse-engineering, and digital-forensics challenges in a jeopardy-style Capture the Flag competition.
- 1st Place (30 teams) - Capture the Flag (CTF) | BSides Puerto Rico | San Juan, PR** **Apr 2024**
- Solved challenges in reverse engineering, cryptography, network analysis, and binary exploitation as part of a 3-member team.
- Selected Student (1 of 29 from 400+) - HSI Tech Summit | Bloomberg | New York, NY** **Apr 2024**
- Selected from 400+ applicants for live coding and engineering sessions; later invited to the 6-week Bloomberg Summer Accelerator.
- 5th Place - Bloomberg Hackathon | UPR Río Piedras | San Juan, PR** **Sep 2022**
- Developed Seniors, a **Python**/PySimpleGUI/SQLite prototype connecting older adults with nearby caregivers or volunteers through location-based requests.
- 2nd Place (team, 40 participants) - Tracer Fire Blue Team Competition | ACSAC (Annual Computer Security Applications Conference)** **Dec 2019**
- Diagnosed compromised accounts and services with **SIEM** and **IDS** tools; performed disk and memory forensics with **Autopsy**, malware analysis with **Wireshark**, and binary reversing with **Ghidra**.

SCHOLARSHIPS

- Francis Castro Scholarship, University of Puerto Rico (2020, 2022)
- Puerto Rico Louis Stokes Alliance for Minority Participation (PR-LSAMP) Undergraduate Fellow (2020)

REFERENCES

Available upon request.